Practical.11

AIM: a Simulate Static Routing configuration using CISCO Packet Tracer.

Static routes are the routes you manually add to the router's routing table. The process of adding static routes to the routing table is known as static routing. Let's take a Packet tracer example to understand how to use static routing to create and add a static route to the routing table.

Setting up a Practice Lab

create a Packet tracer

creating, adding, verifying static routes

Routers automatically learn their connected network. We only need to add routes for the networks that are not available on the router's interfaces. For example, network 10.0.0.0/8, 20.0.0.0/8 and 40.0.0.0/8 are directly

The following table lists the connected networks of each router:

Router	available networks on local interfaces	Networks available on other router interfaces.

Router0

Physical Config CLI Attributes

IOS Command Line Inte

```
Router>en
Router#conf term
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#do show ip interface
GigabitEthernet0/0 is administratively down, line protocol is down (disabled)
  Internet protocol processing disabled
GigabitEthernet0/1 is administratively down, line protocol is down (disabled)
  Internet protocol processing disabled
GigabitEthernet0/2 is administratively down, line protocol is down (disabled)
  Internet protocol processing disabled
Vlan1 is administratively down, line protocol is down
  Internet protocol processing disabled

Router(config)#int g0/0
Router(config-if)#ip address 192.168.1.1 255.255.2553
                                          ^
% Invalid input detected at '^' marker.

Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#int g0/1
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
```

Router1

Physical Config CLI Attributes

IOS Command Line Interf

```
Router>en
Router#conf term
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#do show ip interface
GigabitEthernet0/0 is administratively down, line protocol is down (disabled)
  Internet protocol processing disabled
GigabitEthernet0/1 is administratively down, line protocol is down (disabled)
  Internet protocol processing disabled
GigabitEthernet0/2 is administratively down, line protocol is down (disabled)
  Internet protocol processing disabled
Vlan1 is administratively down, line protocol is down
  Internet protocol processing disabled

Router(config)#int g0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

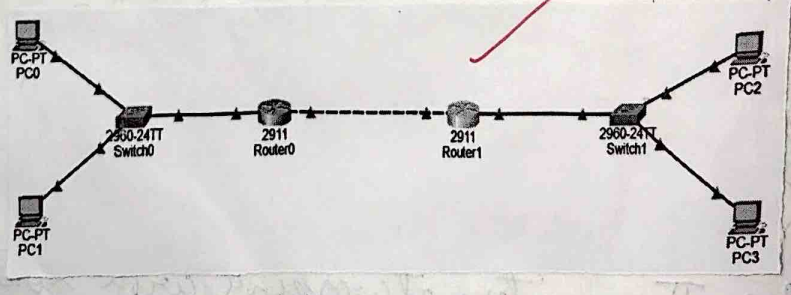
Router(config-if)#exit
Router(config)#interface g0/1
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#
```



Router 0	10.0.0.0/8, 20.0.0.0/8, 40.0.0.0/8,	30.0.0.0/8, 50.0.0.0/8
Router 1	20.0.0.0/8, 30.0.0.0/8, 50.0.0.0/8	10.0.0.0/8, 40.0.0.0/8
Router 2	40.0.0.0/8, 50.0.0.0/8	10.0.0.0/8, 20.0.0.0/8, 30.0.0.0/8

Router0 configuration

Router > enable
Router # configure terminal
Router(config) # ip route 30.0.0.0 255.0.0.0 20.0.0.210
Router(config) # ip route 30.0.0.0 255.0.0.0 40.0.0.220
Router (config) # ip route 30.0.0.100 255.255.255.255 40.0.0.210
Router (config) # ip route 30.0.0.100 255.255.255.255 20.0.0.220
Router (config) # ip route 50.0.0.0 255.0.0.0 40.0.0.210
Router (config) # ip route 50.0.0.0 255.0.0.0 20.0.0.220
Router (config) # exit
Router # show ip route Static

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#copy run-conf startup-conf
                 ^
% Invalid input detected at '^' marker.

Router#copy run-config startup-config
                 ^
% Invalid input detected at '^' marker.

Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/1
L       10.0.0.2/32 is directly connected, GigabitEthernet0/1
S    192.168.1.0/24 [1/0] via 10.0.0.1
     192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/0
L       192.168.2.1/32 is directly connected, GigabitEthernet0/0

Router#
```

30.0.0.0/8 is variably Subnetted, 2 Subnets, 2 masks
S 30.0.0.0/8 [10/0] via 20.0.0.2
S 30.0.0.100/32 [10/0] via

Router 1 Configuration.

Router > enable
Router # configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router (config) # ip route 10.0.0.0 255.0.0.0 20.0.0.1 10
Router (config) # ip route 10.0.0.0 255.0.0.0 50.0.0.1 10
Router (config) # ip route 40.0.0.0 255.0.0.0 20.0.0.1 10
Router (config) # ip route 40.0.0.0 255.0.0.0 50.0.0.1 20
Router (config) # exit
Router # show ip route static
S 10.0.0.0/8 [10/0] via

Router 2 configuration
Router > enable
Router # configure terminal
Enter configuration commands, one per line. End with CNTL/Z

Router(config)# ip route 10.0.0.0 255.
0.0.0 40.0.0.1

Router(config)# ip route 30.0.0.0 255.0.0.0
50.0.0.2 Router(config)# exit

Router# show ip route static

S 10.0.0.0/8 [1/0] via
40.0.0.1 S 30.0.0.0/8 [1/0]

Verfying Static routing

On Router0, we configured two routes for network 30.0.0.0/8. These routes are via Router1 and via Router1 and via Router2. We set the first route (via-Router) as the main route and the second route as the backup route.

By sending Ping requests to a PC of network 30.0.0.0/8 and tracing the Path they you can use "tracert" command on a PC of network 10.0.0.0/8
The 'tracert' command sends Ping requests to the destination host and tracks the Path they take to reach the destination

Deleting a Static route

To delete a static route, use the followings steps.

- Use the 'Show ip route static' Command to print all static routes.
- Note down the route you want to delete.
- Use the "no ip route" command to delete the route. If you have a backup route, the backup route becomes the main route when you delete the main route.

Practical 11

AIM:-b) Simulate RIP using CISCO Packet Tracer

Initial IP configuration

Device	Interface	IP configuration	connected with
PC0	Fast Ethernet	10.0.0.2/8	Router 0's Fa 0/1
Router 0	Fa 0/1	10.0.0.1/8	PC0's Fast Ethernet
Router 0	S0/0/1	192.168.1.254/30	Router 2's S0/0/1
Router 0	S0/0/0	192.168.1.249/30	Router 1's S0/0/0
Router 1	S0/0/0	192.168.1.250/30	Router 0's S0/0/0
Router 1	S0/0/1	192.168.1.246/30	Router 2's S0/0/0
Router 2	S0/0/0	192.168.1.245/30	Router 1's S0/0/1
Router 2	S0/0/1	192.168.1.253/30	Router 0's S0/0/1
Router 2	Fa 0/1	20.0.0.1/30	PC1's Fast Ethernet
PC1	Fast Ethernet	20.0.0.2/30	Router 2's Fa 0/1

Assign IP address to PCs

Double click PCs and click Desktop menu ites and click IP configuration Assign IP address referring the

Router4

Physical Config CLI Attributes

IOS Command Line Inter

```
Press RETURN to get started!


Router>en
Router#conf term
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#do show ip interface
GigabitEthernet0/0 is administratively down, line protocol is down (disabled)
  Internet protocol processing disabled
GigabitEthernet0/1 is administratively down, line protocol is down (disabled)
  Internet protocol processing disabled
GigabitEthernet0/2 is administratively down, line protocol is down (disabled)
  Internet protocol processing disabled
Vlan1 is administratively down, line protocol is down
  Internet protocol processing disabled

Router(config)#interface g0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#route rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 192.168.1.0
Router(config-router)#network 10.0.0.0
Router(config-router)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#
```

Router3

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>en
Router#conf term
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#do show ip interface
GigabitEthernet0/0 is administratively down, line protocol is down (disabled)
  Internet protocol processing disabled
GigabitEthernet0/1 is administratively down, line protocol is down (disabled)
  Internet protocol processing disabled
GigabitEthernet0/2 is administratively down, line protocol is down (disabled)
  Internet protocol processing disabled
Vlan1 is administratively down, line protocol is down
  Internet protocol processing disabled

Router(config)#interface g0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface g0/1
Router(config-if)#ip address 10.0.0.2 255.255.255.253
Bad mask 0xFFFFFFFD for address 10.0.0.2
Router(config-if)#
Router(config-if)#no shutdown

Router(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to down
exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#router rip
          ^
% Invalid input detected at '^' marker.

Router#version 2
          ^
```

Assign IP address to interfaces of routers

Double click Router0 and click CLI and Press Enter key to access the command Prompt of Router0.

Router > enable
Router # configure
terminal
Router (config) # interface fastEthernet 0/0
Router (config-if) # ip address 10.0.0.1 255.0.0.0 Router (config-if) # no Shutdown
Router (config-if) # exit
Router (config) #

interface fastEthernet 0/0 command is used to enter in interface mode.

ip address 10.0.0.1 255.0.0.0 command will assign IP address to interface

no Shutdown command will bring the Interface up.

Router # show controllers Serial 0/0/0
Interface Serial 0/0/0
Hardware is Power QUICC MPC860

```
Router#conf term
Enter configuration commands, one per line.  End with CNTL/Z
Router(config)#router rip
Router(config-router)#ver 2
Router(config-router)#network 192.168.2.0
Router(config-router)#network 20.0.0.0
Router(config-router)#exit
Router(config)#copy running-config startup-config

% Invalid input detected at '^' marker.

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

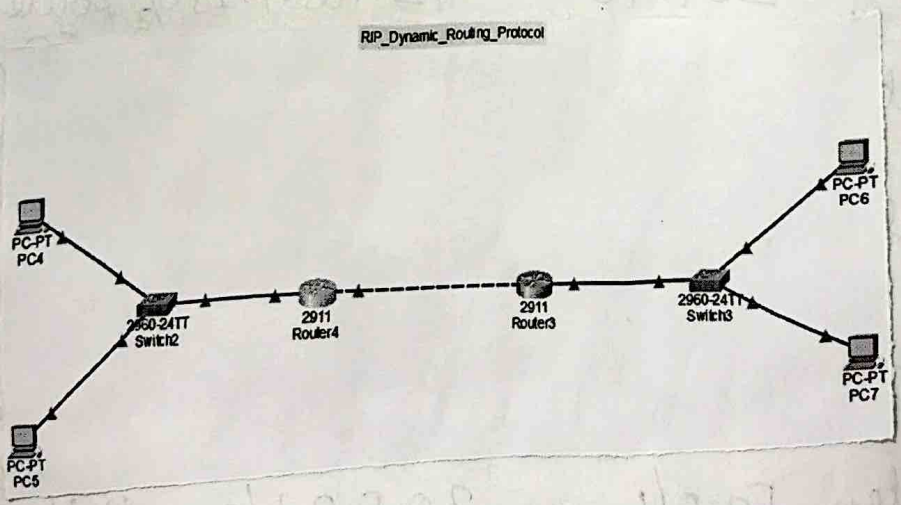
Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#
Router#
```

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/0
L       192.168.2.1/32 is directly connected, GigabitEthernet0/0

Router#
```



DCE V.35, Clock rate 2000000

[output omitted]

Router # configure terminal

Enter configuration commands, one per line
End with CNTL/Z

Router (config) # interface Serial 0/0/0

Router (config) # IP address 192.168.1.249
255.255.255.252

Router (config) # clock rate 64000

Router (config) # bandwidth 64

Router (config-if) # no shutdown

Router (config-if) # exit

Router (config) # interface serial 0/0/1

Router (config-if) # clock rate 64000

Router (config-if) # bandwidth
64 Router (config-if) # no
Shutdown Router (config)

Router # configure terminal

Router (config) # interface Serial 0/0/0

Router (config-if) # IP address (192.168.1.
249 255.255.255.252

Router (config-if) # clock rate 64000

Router (config-if) # no Shutdown

Router (Config-if) # exit

Router1

Router > enable

Router # configure

terminal

Enter configuration commands. one

Per line. End with CNTL/Z

Router (config) # interface Serial 0/0/0

Router (config-if) # no Shutdown

Router (config-if) # exit

Router (config) # interface Serial 0/0/1

Router (config-if) # ip address 192.168.1.

246. 255. 255. 255. 252

Router (config-if) # clock rate 64000

Router (config-if) # band width

64 Router (config-if) # no

Shutdown Router (config-)

Router2

Router > enable

Router # configure

terminal

Enter configuration commands. one

Per line End with CNTL/Z

Router(config)# interface fastEthernet 0/0
Router(config-if)# ip address 20.0.0.1
255.0.0.0 Router(config-if)# no Shutdown
Router(config-if)# exit
Router(config)# interface serial 0/0/0
Router(config-if)# ip address 192.168.1.245
255.255.255.252
Router(config-if)# no Shutdown.
Router(config)# exit
Router(config)# interface Serial 0/0/1
Router(config)# ip address 192.168.1.253

Configure RIP Routing Protocol

- Enable RIP routing protocol from global configuration mode.
- Tell RIP routing protocol with networks you wat to advertise. Lets Configure it in Router

Router 0(config)# router rip
Router 0(config-router)# network
10.0.0.0
Router(config-router)# network
192.168.1.252

Router 0

router rip command tell router to enable the RIP routing Protocol. network command allows us to Specify the networks which we want to advertise

Router 1

Router 1 (config) # router rip

Router 1 (config-router) # network 192.168.1.244

Router 1 (config-router) # network 192.168.1.248

Router 2

Router 2 (config) # router rip

Router 2 (config-router) # network 20.0.0.0

Router 2 (config-router) # network 192.168.1.252

Router 2 (config-router) # network 192.168.1.244

Result :-

This experiment implements the Routing at Network layer using both Static and Dynamic Routing.